Volume

Astra Automation

NetApp
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Volume

Volume overview

The Volume API provides visibility into Kubernetes volumes associated with applications.

List the volumes for a managed cluster

GET
`/accounts/{account_id}/topology/v1/managedClusters/{managedCluster_id}/volumes`

Returns a JSON array of all matching resources in the collection. If individual fields are specified in the request, the items JSON array will contain the fields requested for each matching resource in the order specified.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}
managedCluster_id	string	path	True	ID of the containing managedCluster resource
include	string	query	False	Indicates which fields should be returned when listing a collection. URL examples: "include=id", "include=id,name" OpenAPI examples: "id" • example: id,name

Name	Type	In	Required	Description
limit	string	query	False	Indicates the maximum number of resources to return when listing a collection. URL examples: "limit=2" OpenAPI examples: "2"

Response

```
Status: 200, Returns a JSON array of all matching resources in the
collection. If individual fields are specified in the request, the items
JSON array will contain the fields requested for each matching resource in
the order specified.
```

Name	Type	Required	Description
items	array[Any of: volume_1.2_get_response_body, array[]]	True	
metadata	type_astra_metadata_list	True	

Example response


```
{
  "type": "application/astra-volumes",
  "version": "1.2",
  "items": [
    [
      "24fe6e79-bf8d-45c9-96af-ef1d11fbd4a0",
      "myVolume"
    ],
    [
      "20a370f4-957f-446a-9126-08853d01933a",
      "secondVol"
    ],
    [
      "48e349a2-5bcd-4c59-896c-12b8240e634b",
      "smallVol"
    ]
  ],
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```


Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	
invalidParams	array[invalidParams]	False	List of invalid query parameters

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't
found.",
  "status": "404"
}
```

Definitions

See Definitions

additionalDetails

type_astra_stateDetail

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
additionalDetails	additionalDetails	False	

type_astra_label

Name	Type	Required	Description
name	string	True	
value	string	True	

type_astra_metadata

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema

Name	Type	Required	Description
labels	array[type_astra_label]	True	
creationTimestamp	string	True	
modificationTimestamp	string	True	
createdBy	string	True	
modifiedBy	string	False	

volume_1.2_get_response_body

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-volume"

Name	Type	Required	Description
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • "1.2"
id	string	True	Globally unique identifier of the resource. Defined values are: • Conforms to the UUIDv4 Schema
name	string	True	JSON string containing the name of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255
state	string	False	JSON string containing a value indicating the operational state of the volume.
size	string	False	JSON string containing the size of the volume.
used	number	False	JSON number containing the number of bytes of quota used by the provisioned volume.
usedPercentage	number	False	JSON number containing the percentage of volume used.
total	number	False	JSON number containing the total bytes of volumes provisioned.

Name	Type	Required	Description
creationToken	string	False	JSON string containing the creation token of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255
snapshotPossible	string	False	JSON string indicating whether the driver can perform a snapshot of the volume. Defined values are: • "true" - Driver can take a snapshot of the volume • "false" - Driver cannot take a snapshot of the volume
storageClass	string	False	JSON string containing the storage class for the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255
pvcName	string	False	JSON string containing the PersistentVolumeClaim (PVC) name of the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255

Name	Type	Required	Description
internalName	string	False	JSON string containing the internal name of the volume on the target storage cluster, if available. This name is obtained from the Kubernetes CSI driver. • Defined values are: • Minimum length: 1 • Maximum length: 255
appsUsing	array[string]	True	JSON array of JSON strings, each containing the ID of an application that is using the volume. Defined values are: • Conforms to the Astra Identifier Schema
storageBackendID	string	False	JSON string containing the ID of the storage backend that provides the volume. Defined values are: • Conforms to the Astra Identifier Schema

Name	Type	Required	Description
orchestrator	string	False	JSON string containing the orchestrator for this volume. This will be defined for Astra Data Store volumes. Defined values are: • "trident" - Trident has provisioned this volume • "vasa" - NetApp VASA Provider has provisioned this volume • "other" - The volume has been provisioned manually or the orchestrator is not known
serviceLevel	string	False	JSON string containing the quality of service (QoS) policy defined for this volume at the storage backend level. This will be defined for Astra Data Store volumes.
exportPolicy	string	False	JSON string containing the export policy associated with this volume. This will be defined for Astra Data Store volumes.
nodeId	string	False	JSON string containing the node ID serving the volume. This will be defined for Astra Data Store volumes. Defined values are: • Conforms to the Astra Identifier Schema

Name	Type	Required	Description
healthState	string	False	JSON string containing a value indicating the health state of the volume. This will be defined for Astra Data Store volumes. Defined values are: • "indeterminate" - The health of the resource cannot be determined (for example, due to lack of connectivity) • "normal" - The resource has no known issues and can provide services in accordance with the interface specification • "warning" - The resource has known issues that do not prevent it providing services in accordance with the interface specification • "critical" - The resource has known issues that prevent it from providing services in accordance with the interface specification
healthStateDetails	array[type_astra_stateDetail]	True	JSON array of JSON objects that lists additional details about the current health. If no details are available, this array will be empty. • Minimum Items: 0

Name	Type	Required	Description
metadata	type_astra_metadata	True	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema

type_astra_metadata_list

Name	Type	Required	Description
continue	string	False	
count	number	False	
labels	array[type_astra_label]	False	
creationTimestamp	string	False	
modificationTimestamp	string	False	
createdBy	string	False	
modifiedBy	string	False	

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

Retrieve a volume for a managed cluster

GET `/accounts/{account_id}/topology/v1/managedClusters/{managedCluster_id}/volumes/{volume_id}`

Returns the stored Volume API resource in the JSON response body.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}
managedCluster_id	string	path	True	ID of the containing managedCluster resource
volume_id	string	path	True	ID of the volume collection to list

Response

```
Status: 200, Returns the stored Volume API resource in the JSON response
body.
```

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-volume"
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • "1.2"
id	string	True	Globally unique identifier of the resource. Defined values are: • Conforms to the UUIDv4 Schema

Name	Type	Required	Description
name	string	True	JSON string containing the name of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255
state	string	False	JSON string containing a value indicating the operational state of the volume.
size	string	False	JSON string containing the size of the volume.
used	number	False	JSON number containing the number of bytes of quota used by the provisioned volume.
usedPercentage	number	False	JSON number containing the percentage of volume used.
total	number	False	JSON number containing the total bytes of volumes provisioned.
creationToken	string	False	JSON string containing the creation token of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255

Name	Type	Required	Description
snapshotPossible	string	False	JSON string indicating whether the driver can perform a snapshot of the volume. Defined values are: • "true" - Driver can take a snapshot of the volume • "false" - Driver cannot take a snapshot of the volume
storageClass	string	False	JSON string containing the storage class for the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255
pvcName	string	False	JSON string containing the PersistentVolumeClaim (PVC) name of the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255
internalName	string	False	JSON string containing the internal name of the volume on the target storage cluster, if available. This name is obtained from the Kubernetes CSI driver. • Defined values are: • Minimum length: 1 • Maximum length: 255

Name	Type	Required	Description
appsUsing	array[string]	True	JSON array of JSON strings, each containing the ID of an application that is using the volume. Defined values are: • Conforms to the Astra Identifier Schema
storageBackendID	string	False	JSON string containing the ID of the storage backend that provides the volume. Defined values are: • Conforms to the Astra Identifier Schema
orchestrator	string	False	JSON string containing the orchestrator for this volume. This will be defined for Astra Data Store volumes. Defined values are: • "trident" - Trident has provisioned this volume • "vasa" - NetApp VASA Provider has provisioned this volume • "other" - The volume has been provisioned manually or the orchestrator is not known
serviceLevel	string	False	JSON string containing the quality of service (QoS) policy defined for this volume at the storage backend level. This will be defined for Astra Data Store volumes.

Name	Type	Required	Description
exportPolicy	string	False	JSON string containing the export policy associated with this volume. This will be defined for Astra Data Store volumes.
nodeId	string	False	JSON string containing the node ID serving the volume. This will be defined for Astra Data Store volumes. Defined values are: • Conforms to the Astra Identifier Schema
healthState	string	False	JSON string containing a value indicating the health state of the volume. This will be defined for Astra Data Store volumes. Defined values are: • "indeterminate" - The health of the resource cannot be determined (for example, due to lack of connectivity) • "normal" - The resource has no known issues and can provide services in accordance with the interface specification • "warning" - The resource has known issues that do not prevent it providing services in accordance with the interface specification • "critical" - The resource has known issues that prevent it from providing services in accordance with the interface specification

Name	Type	Required	Description
healthStateDetails	array[type_astra_stateDetail]	True	JSON array of JSON objects that lists additional details about the current health. If no details are available, this array will be empty. • Minimum Items: 0
metadata	type_astra_metadata	True	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema

Example response


```
{
  "type": "application/astra-volume",
  "version": "1.2",
  "id": "ab5c9fb8-ddba-526f-8bd3-8d5b72cf4f7f",
  "name": "myVolume",
  "state": "ready",
  "size": "512 GiB",
  "used": 10000000000,
  "usedPercentage": 10,
  "total": 10000000000,
  "creationToken": "ba7d4e0c-6ae3-4b6e-859d-9463e7ca4b4e",
  "snapshotPossible": "true",
  "storageClass": "gold",
  "pvcName": "myVolume",
  "internalName": "trident_sd234234_23424_897d_876sd65w",
  "appsUsing": [],
  "orchestrator": "trident",
  "serviceLevel": "gold",
  "exportPolicy": "astra-vp-default",
  "nodeId": "94d16dd2-9beb-4edb-ab58-1a00e9a90052",
  "healthState": "normal",
  "healthStateDetails": [],
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```


Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	

Name	Type	Required	Description
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	
invalidParams	array[invalidParams]	False	List of invalid query parameters

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't
found.",
  "status": "404"
}
```

Definitions

See Definitions

additionalDetails

type_astra_stateDetail

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
additionalDetails	additionalDetails	False	

type_astra_label

Name	Type	Required	Description
name	string	True	
value	string	True	

type_astra_metadata

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema

Name	Type	Required	Description
labels	array[type_astra_label]	True	
creationTimestamp	string	True	
modificationTimestamp	string	True	
createdBy	string	True	
modifiedBy	string	False	

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

List the volumes for a storage backend

GET
`/accounts/{account_id}/topology/v1/storageBackends/{storageBackend_id}/volumes`

Returns a JSON array of all matching resources in the collection. If individual fields are specified in the request, the items JSON array will contain the fields requested for each matching resource in the order specified.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}
storageBackend_id	string	path	True	ID of the containing storageBackend resource
include	string	query	False	Indicates which fields should be returned when listing a collection. URL examples: "include=id", "include=id,name" OpenAPI examples: "id" • example: id,name
limit	string	query	False	Indicates the maximum number of resources to return when listing a collection. URL examples: "limit=2" OpenAPI examples: "2"

Response

```
Status: 200, Returns a JSON array of all matching resources in the
collection. If individual fields are specified in the request, the items
JSON array will contain the fields requested for each matching resource in
the order specified.
```

Name	Type	Required	Description
items	array[Any of: volume_1.2_get_response_body, array[]]	True	
metadata	type_astra_metadata_list	True	

Example response


```
{
  "type": "application/astra-volumes",
  "version": "1.2",
  "items": [
    [
      "24fe6e79-bf8d-45c9-96af-ef1d11fbd4a0",
      "myVolume"
    ],
    [
      "20a370f4-957f-446a-9126-08853d01933a",
      "secondVol"
    ],
    [
      "48e349a2-5bcd-4c59-896c-12b8240e634b",
      "smallVol"
    ]
  ],
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```


Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	
invalidParams	array[invalidParams]	False	List of invalid query parameters

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	

Name	Type	Required	Description
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't
found.",
  "status": "404"
}
```

Definitions

See Definitions

additionalDetails

type_astra_stateDetail

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
additionalDetails	additionalDetails	False	

type_astra_label

Name	Type	Required	Description
name	string	True	
value	string	True	

type_astra_metadata

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema

Name	Type	Required	Description
labels	array[type_astra_label]	True	
creationTimestamp	string	True	
modificationTimestamp	string	True	
createdBy	string	True	
modifiedBy	string	False	

volume_1.2_get_response_body

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-volume"

Name	Type	Required	Description
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • "1.2"
id	string	True	Globally unique identifier of the resource. Defined values are: • Conforms to the UUIDv4 Schema
name	string	True	JSON string containing the name of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255
state	string	False	JSON string containing a value indicating the operational state of the volume.
size	string	False	JSON string containing the size of the volume.
used	number	False	JSON number containing the number of bytes of quota used by the provisioned volume.
usedPercentage	number	False	JSON number containing the percentage of volume used.
total	number	False	JSON number containing the total bytes of volumes provisioned.

Name	Type	Required	Description
creationToken	string	False	JSON string containing the creation token of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255
snapshotPossible	string	False	JSON string indicating whether the driver can perform a snapshot of the volume. Defined values are: • "true" - Driver can take a snapshot of the volume • "false" - Driver cannot take a snapshot of the volume
storageClass	string	False	JSON string containing the storage class for the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255
pvcName	string	False	JSON string containing the PersistentVolumeClaim (PVC) name of the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255

Name	Type	Required	Description
internalName	string	False	JSON string containing the internal name of the volume on the target storage cluster, if available. This name is obtained from the Kubernetes CSI driver. • Defined values are: • Minimum length: 1 • Maximum length: 255
appsUsing	array[string]	True	JSON array of JSON strings, each containing the ID of an application that is using the volume. Defined values are: • Conforms to the Astra Identifier Schema
storageBackendID	string	False	JSON string containing the ID of the storage backend that provides the volume. Defined values are: • Conforms to the Astra Identifier Schema

Name	Type	Required	Description
orchestrator	string	False	JSON string containing the orchestrator for this volume. This will be defined for Astra Data Store volumes. Defined values are: • "trident" - Trident has provisioned this volume • "vasa" - NetApp VASA Provider has provisioned this volume • "other" - The volume has been provisioned manually or the orchestrator is not known
serviceLevel	string	False	JSON string containing the quality of service (QoS) policy defined for this volume at the storage backend level. This will be defined for Astra Data Store volumes.
exportPolicy	string	False	JSON string containing the export policy associated with this volume. This will be defined for Astra Data Store volumes.
nodeId	string	False	JSON string containing the node ID serving the volume. This will be defined for Astra Data Store volumes. Defined values are: • Conforms to the Astra Identifier Schema

Name	Type	Required	Description
healthState	string	False	JSON string containing a value indicating the health state of the volume. This will be defined for Astra Data Store volumes. Defined values are: • "indeterminate" - The health of the resource cannot be determined (for example, due to lack of connectivity) • "normal" - The resource has no known issues and can provide services in accordance with the interface specification • "warning" - The resource has known issues that do not prevent it providing services in accordance with the interface specification • "critical" - The resource has known issues that prevent it from providing services in accordance with the interface specification
healthStateDetails	array[type_astra_stateDetail]	True	JSON array of JSON objects that lists additional details about the current health. If no details are available, this array will be empty. • Minimum Items: 0

Name	Type	Required	Description
metadata	type_astra_metadata	True	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema

type_astra_metadata_list

Name	Type	Required	Description
continue	string	False	
count	number	False	
labels	array[type_astra_label]	False	
creationTimestamp	string	False	
modificationTimestamp	string	False	
createdBy	string	False	
modifiedBy	string	False	

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

Retrieve a volume for a storage backend

GET `/accounts/{account_id}/topology/v1/storageBackends/{storageBackend_id}/volumes/{volume_id}`

Returns the stored Volume API resource in the JSON response body.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}
storageBackend_id	string	path	True	ID of the containing storageBackend resource
volume_id	string	path	True	ID of the volume collection to list

Response

```
Status: 200, Returns the stored Volume API resource in the JSON response
body.
```

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-volume"
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • "1.2"
id	string	True	Globally unique identifier of the resource. Defined values are: • Conforms to the UUIDv4 Schema

Name	Type	Required	Description
name	string	True	JSON string containing the name of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255
state	string	False	JSON string containing a value indicating the operational state of the volume.
size	string	False	JSON string containing the size of the volume.
used	number	False	JSON number containing the number of bytes of quota used by the provisioned volume.
usedPercentage	number	False	JSON number containing the percentage of volume used.
total	number	False	JSON number containing the total bytes of volumes provisioned.
creationToken	string	False	JSON string containing the creation token of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255

Name	Type	Required	Description
snapshotPossible	string	False	JSON string indicating whether the driver can perform a snapshot of the volume. Defined values are: • "true" - Driver can take a snapshot of the volume • "false" - Driver cannot take a snapshot of the volume
storageClass	string	False	JSON string containing the storage class for the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255
pvcName	string	False	JSON string containing the PersistentVolumeClaim (PVC) name of the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255
internalName	string	False	JSON string containing the internal name of the volume on the target storage cluster, if available. This name is obtained from the Kubernetes CSI driver. • Defined values are: • Minimum length: 1 • Maximum length: 255

Name	Type	Required	Description
appsUsing	array[string]	True	JSON array of JSON strings, each containing the ID of an application that is using the volume. Defined values are: • Conforms to the Astra Identifier Schema
storageBackendID	string	False	JSON string containing the ID of the storage backend that provides the volume. Defined values are: • Conforms to the Astra Identifier Schema
orchestrator	string	False	JSON string containing the orchestrator for this volume. This will be defined for Astra Data Store volumes. Defined values are: • "trident" - Trident has provisioned this volume • "vasa" - NetApp VASA Provider has provisioned this volume • "other" - The volume has been provisioned manually or the orchestrator is not known
serviceLevel	string	False	JSON string containing the quality of service (QoS) policy defined for this volume at the storage backend level. This will be defined for Astra Data Store volumes.

Name	Type	Required	Description
exportPolicy	string	False	JSON string containing the export policy associated with this volume. This will be defined for Astra Data Store volumes.
nodeId	string	False	JSON string containing the node ID serving the volume. This will be defined for Astra Data Store volumes. Defined values are: • Conforms to the Astra Identifier Schema
healthState	string	False	JSON string containing a value indicating the health state of the volume. This will be defined for Astra Data Store volumes. Defined values are: • "indeterminate" - The health of the resource cannot be determined (for example, due to lack of connectivity) • "normal" - The resource has no known issues and can provide services in accordance with the interface specification • "warning" - The resource has known issues that do not prevent it providing services in accordance with the interface specification • "critical" - The resource has known issues that prevent it from providing services in accordance with the interface specification

Name	Type	Required	Description
healthStateDetails	array[type_astra_stateDetail]	True	JSON array of JSON objects that lists additional details about the current health. If no details are available, this array will be empty. • Minimum Items: 0
metadata	type_astra_metadata	True	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema

Example response


```
{
  "type": "application/astra-volume",
  "version": "1.2",
  "id": "ab5c9fb8-ddba-526f-8bd3-8d5b72cf4f7f",
  "name": "myVolume",
  "state": "ready",
  "size": "512 GiB",
  "used": 10000000000,
  "usedPercentage": 10,
  "total": 10000000000,
  "creationToken": "ba7d4e0c-6ae3-4b6e-859d-9463e7ca4b4e",
  "snapshotPossible": "true",
  "storageClass": "gold",
  "pvcName": "myVolume",
  "internalName": "trident_sd234234_23424_897d_876sd65w",
  "appsUsing": [],
  "orchestrator": "trident",
  "serviceLevel": "gold",
  "exportPolicy": "astra-vp-default",
  "nodeId": "94d16dd2-9beb-4edb-ab58-1a00e9a90052",
  "healthState": "normal",
  "healthStateDetails": [],
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```


Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	

Name	Type	Required	Description
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	
invalidParams	array[invalidParams]	False	List of invalid query parameters

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't
found.",
  "status": "404"
}
```

Definitions

See Definitions

additionalDetails

type_astra_stateDetail

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
additionalDetails	additionalDetails	False	

type_astra_label

Name	Type	Required	Description
name	string	True	
value	string	True	

type_astra_metadata

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema

Name	Type	Required	Description
labels	array[type_astra_label]	True	
creationTimestamp	string	True	
modificationTimestamp	string	True	
createdBy	string	True	
modifiedBy	string	False	

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

List all volumes

GET `/accounts/{account_id}/topology/v1/volumes`

Returns a JSON array of all matching resources in the collection. If individual fields are specified in the request, the items JSON array will contain the fields requested for each matching resource in the order specified.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}
include	string	query	False	Indicates which fields should be returned when listing a collection. URL examples: "include=id", "include=id,name" OpenAPI examples: "id" • example: id,name
limit	string	query	False	Indicates the maximum number of resources to return when listing a collection. URL examples: "limit=2" OpenAPI examples: "2"

Response

```
Status: 200, Returns a JSON array of all matching resources in the
collection. If individual fields are specified in the request, the items
JSON array will contain the fields requested for each matching resource in
the order specified.
```

Name	Type	Required	Description
items	array[Any of: volume_1.2_get_response_body, array[]]	True	
metadata	type_astra_metadata_list	True	

Example response


```
{
  "type": "application/astra-volumes",
  "version": "1.2",
  "items": [
    [
      "24fe6e79-bf8d-45c9-96af-ef1d11fbd4a0",
      "myVolume"
    ],
    [
      "20a370f4-957f-446a-9126-08853d01933a",
      "secondVol"
    ],
    [
      "48e349a2-5bcd-4c59-896c-12b8240e634b",
      "smallVol"
    ]
  ],
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```


Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	
invalidParams	array[invalidParams]	False	List of invalid query parameters

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	

Name	Type	Required	Description
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't
found.",
  "status": "404"
}
```

Definitions

See Definitions

additionalDetails

type_astra_stateDetail

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
additionalDetails	additionalDetails	False	

type_astra_label

Name	Type	Required	Description
name	string	True	
value	string	True	

type_astra_metadata

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema

Name	Type	Required	Description
labels	array[type_astra_label]	True	
creationTimestamp	string	True	
modificationTimestamp	string	True	
createdBy	string	True	
modifiedBy	string	False	

volume_1.2_get_response_body

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-volume"

Name	Type	Required	Description
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • "1.2"
id	string	True	Globally unique identifier of the resource. Defined values are: • Conforms to the UUIDv4 Schema
name	string	True	JSON string containing the name of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255
state	string	False	JSON string containing a value indicating the operational state of the volume.
size	string	False	JSON string containing the size of the volume.
used	number	False	JSON number containing the number of bytes of quota used by the provisioned volume.
usedPercentage	number	False	JSON number containing the percentage of volume used.
total	number	False	JSON number containing the total bytes of volumes provisioned.

Name	Type	Required	Description
creationToken	string	False	JSON string containing the creation token of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255
snapshotPossible	string	False	JSON string indicating whether the driver can perform a snapshot of the volume. Defined values are: • "true" - Driver can take a snapshot of the volume • "false" - Driver cannot take a snapshot of the volume
storageClass	string	False	JSON string containing the storage class for the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255
pvcName	string	False	JSON string containing the PersistentVolumeClaim (PVC) name of the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255

Name	Type	Required	Description
internalName	string	False	JSON string containing the internal name of the volume on the target storage cluster, if available. This name is obtained from the Kubernetes CSI driver. • Defined values are: • Minimum length: 1 • Maximum length: 255
appsUsing	array[string]	True	JSON array of JSON strings, each containing the ID of an application that is using the volume. Defined values are: • Conforms to the Astra Identifier Schema
storageBackendID	string	False	JSON string containing the ID of the storage backend that provides the volume. Defined values are: • Conforms to the Astra Identifier Schema

Name	Type	Required	Description
orchestrator	string	False	JSON string containing the orchestrator for this volume. This will be defined for Astra Data Store volumes. Defined values are: • "trident" - Trident has provisioned this volume • "vasa" - NetApp VASA Provider has provisioned this volume • "other" - The volume has been provisioned manually or the orchestrator is not known
serviceLevel	string	False	JSON string containing the quality of service (QoS) policy defined for this volume at the storage backend level. This will be defined for Astra Data Store volumes.
exportPolicy	string	False	JSON string containing the export policy associated with this volume. This will be defined for Astra Data Store volumes.
nodeId	string	False	JSON string containing the node ID serving the volume. This will be defined for Astra Data Store volumes. Defined values are: • Conforms to the Astra Identifier Schema

Name	Type	Required	Description
healthState	string	False	JSON string containing a value indicating the health state of the volume. This will be defined for Astra Data Store volumes. Defined values are: • "indeterminate" - The health of the resource cannot be determined (for example, due to lack of connectivity) • "normal" - The resource has no known issues and can provide services in accordance with the interface specification • "warning" - The resource has known issues that do not prevent it providing services in accordance with the interface specification • "critical" - The resource has known issues that prevent it from providing services in accordance with the interface specification
healthStateDetails	array[type_astra_stateDetail]	True	JSON array of JSON objects that lists additional details about the current health. If no details are available, this array will be empty. • Minimum Items: 0

Name	Type	Required	Description
metadata	type_astra_metadata	True	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema

type_astra_metadata_list

Name	Type	Required	Description
continue	string	False	
count	number	False	
labels	array[type_astra_label]	False	
creationTimestamp	string	False	
modificationTimestamp	string	False	
createdBy	string	False	
modifiedBy	string	False	

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

Retrieve a volume

GET `/accounts/{account_id}/topology/v1/volumes/{volume_id}`

Returns the stored Volume API resource in the JSON response body.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}
volume_id	string	path	True	ID of the volume collection to list

Response

```
Status: 200, Returns the stored Volume API resource in the JSON response
body.
```

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-volume"
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • "1.2"
id	string	True	Globally unique identifier of the resource. Defined values are: • Conforms to the UUIDv4 Schema
name	string	True	JSON string containing the name of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255

Name	Type	Required	Description
state	string	False	JSON string containing a value indicating the operational state of the volume.
size	string	False	JSON string containing the size of the volume.
used	number	False	JSON number containing the number of bytes of quota used by the provisioned volume.
usedPercentage	number	False	JSON number containing the percentage of volume used.
total	number	False	JSON number containing the total bytes of volumes provisioned.
creationToken	string	False	JSON string containing the creation token of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255
snapshotPossible	string	False	JSON string indicating whether the driver can perform a snapshot of the volume. Defined values are: • "true" - Driver can take a snapshot of the volume • "false" - Driver cannot take a snapshot of the volume

Name	Type	Required	Description
storageClass	string	False	JSON string containing the storage class for the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255
pvcName	string	False	JSON string containing the PersistentVolumeClaim (PVC) name of the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255
internalName	string	False	JSON string containing the internal name of the volume on the target storage cluster, if available. This name is obtained from the Kubernetes CSI driver. • Defined values are: • Minimum length: 1 • Maximum length: 255
appsUsing	array[string]	True	JSON array of JSON strings, each containing the ID of an application that is using the volume. Defined values are: • Conforms to the Astra Identifier Schema
storageBackendID	string	False	JSON string containing the ID of the storage backend that provides the volume. Defined values are: • Conforms to the Astra Identifier Schema

Name	Type	Required	Description
orchestrator	string	False	JSON string containing the orchestrator for this volume. This will be defined for Astra Data Store volumes. Defined values are: • "trident" - Trident has provisioned this volume • "vasa" - NetApp VASA Provider has provisioned this volume • "other" - The volume has been provisioned manually or the orchestrator is not known
serviceLevel	string	False	JSON string containing the quality of service (QoS) policy defined for this volume at the storage backend level. This will be defined for Astra Data Store volumes.
exportPolicy	string	False	JSON string containing the export policy associated with this volume. This will be defined for Astra Data Store volumes.
nodeId	string	False	JSON string containing the node ID serving the volume. This will be defined for Astra Data Store volumes. Defined values are: • Conforms to the Astra Identifier Schema

Name	Type	Required	Description
healthState	string	False	JSON string containing a value indicating the health state of the volume. This will be defined for Astra Data Store volumes. Defined values are: • "indeterminate" - The health of the resource cannot be determined (for example, due to lack of connectivity) • "normal" - The resource has no known issues and can provide services in accordance with the interface specification • "warning" - The resource has known issues that do not prevent it providing services in accordance with the interface specification • "critical" - The resource has known issues that prevent it from providing services in accordance with the interface specification
healthStateDetails	array[type_astra_stateDetail]	True	JSON array of JSON objects that lists additional details about the current health. If no details are available, this array will be empty. • Minimum Items: 0

Name	Type	Required	Description
metadata	type_astra_metadata	True	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema

Example response


```
{
  "type": "application/astra-volume",
  "version": "1.2",
  "id": "ab5c9fb8-ddba-526f-8bd3-8d5b72cf4f7f",
  "name": "myVolume",
  "state": "ready",
  "size": "512 GiB",
  "used": 10000000000,
  "usedPercentage": 10,
  "total": 10000000000,
  "creationToken": "ba7d4e0c-6ae3-4b6e-859d-9463e7ca4b4e",
  "snapshotPossible": "true",
  "storageClass": "gold",
  "pvcName": "myVolume",
  "internalName": "trident_sd234234_23424_897d_876sd65w",
  "appsUsing": [],
  "orchestrator": "trident",
  "serviceLevel": "gold",
  "exportPolicy": "astra-vp-default",
  "nodeId": "94d16dd2-9beb-4edb-ab58-1a00e9a90052",
  "healthState": "normal",
  "healthStateDetails": [],
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```

Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	
invalidParams	array[invalidParams]	False	List of invalid query parameters

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	

Name	Type	Required	Description
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't
found.",
  "status": "404"
}
```

Definitions

See Definitions

additionalDetails

type_astra_stateDetail

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
additionalDetails	additionalDetails	False	

type_astra_label

Name	Type	Required	Description
name	string	True	
value	string	True	

type_astra_metadata

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema

Name	Type	Required	Description
labels	array[type_astra_label]	True	
creationTimestamp	string	True	
modificationTimestamp	string	True	
createdBy	string	True	
modifiedBy	string	False	

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

List the volumes for an app

GET `/accounts/{account_id}/k8s/v1/apps/{app_id}/volumes`

Returns a JSON array of all matching resources in the collection. If individual fields are specified in the request, the items JSON array will contain the fields requested for each matching resource in the order specified.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}
app_id	string	path	True	ID of the containing app resource
include	string	query	False	Indicates which fields should be returned when listing a collection. URL examples: "include=id", "include=id,name" OpenAPI examples: "id" • example: id,name
limit	string	query	False	Indicates the maximum number of resources to return when listing a collection. URL examples: "limit=2" OpenAPI examples: "2"

Response

```
Status: 200, Returns a JSON array of all matching resources in the
collection. If individual fields are specified in the request, the items
JSON array will contain the fields requested for each matching resource in
the order specified.
```

Name	Type	Required	Description
items	array[Any of: volume_1.2_get_response_body, array[]]	True	
metadata	type_astra_metadata_list	True	

Example response


```
{
  "type": "application/astra-volumes",
  "version": "1.2",
  "items": [
    [
      "24fe6e79-bf8d-45c9-96af-ef1d11fbd4a0",
      "myVolume"
    ],
    [
      "20a370f4-957f-446a-9126-08853d01933a",
      "secondVol"
    ],
    [
      "48e349a2-5bcd-4c59-896c-12b8240e634b",
      "smallVol"
    ]
  ],
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```


Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	
invalidParams	array[invalidParams]	False	List of invalid query parameters

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	

Name	Type	Required	Description
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't
found.",
  "status": "404"
}
```

Definitions

See Definitions

additionalDetails

type_astra_stateDetail

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
additionalDetails	additionalDetails	False	

type_astra_label

Name	Type	Required	Description
name	string	True	
value	string	True	

type_astra_metadata

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema

Name	Type	Required	Description
labels	array[type_astra_label]	True	
creationTimestamp	string	True	
modificationTimestamp	string	True	
createdBy	string	True	
modifiedBy	string	False	

volume_1.2_get_response_body

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-volume"

Name	Type	Required	Description
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • "1.2"
id	string	True	Globally unique identifier of the resource. Defined values are: • Conforms to the UUIDv4 Schema
name	string	True	JSON string containing the name of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255
state	string	False	JSON string containing a value indicating the operational state of the volume.
size	string	False	JSON string containing the size of the volume.
used	number	False	JSON number containing the number of bytes of quota used by the provisioned volume.
usedPercentage	number	False	JSON number containing the percentage of volume used.
total	number	False	JSON number containing the total bytes of volumes provisioned.

Name	Type	Required	Description
creationToken	string	False	JSON string containing the creation token of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255
snapshotPossible	string	False	JSON string indicating whether the driver can perform a snapshot of the volume. Defined values are: • "true" - Driver can take a snapshot of the volume • "false" - Driver cannot take a snapshot of the volume
storageClass	string	False	JSON string containing the storage class for the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255
pvcName	string	False	JSON string containing the PersistentVolumeClaim (PVC) name of the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255

Name	Type	Required	Description
internalName	string	False	JSON string containing the internal name of the volume on the target storage cluster, if available. This name is obtained from the Kubernetes CSI driver. • Defined values are: • Minimum length: 1 • Maximum length: 255
appsUsing	array[string]	True	JSON array of JSON strings, each containing the ID of an application that is using the volume. Defined values are: • Conforms to the Astra Identifier Schema
storageBackendID	string	False	JSON string containing the ID of the storage backend that provides the volume. Defined values are: • Conforms to the Astra Identifier Schema

Name	Type	Required	Description
orchestrator	string	False	JSON string containing the orchestrator for this volume. This will be defined for Astra Data Store volumes. Defined values are: • "trident" - Trident has provisioned this volume • "vasa" - NetApp VASA Provider has provisioned this volume • "other" - The volume has been provisioned manually or the orchestrator is not known
serviceLevel	string	False	JSON string containing the quality of service (QoS) policy defined for this volume at the storage backend level. This will be defined for Astra Data Store volumes.
exportPolicy	string	False	JSON string containing the export policy associated with this volume. This will be defined for Astra Data Store volumes.
nodeId	string	False	JSON string containing the node ID serving the volume. This will be defined for Astra Data Store volumes. Defined values are: • Conforms to the Astra Identifier Schema

Name	Type	Required	Description
healthState	string	False	JSON string containing a value indicating the health state of the volume. This will be defined for Astra Data Store volumes. Defined values are: • "indeterminate" - The health of the resource cannot be determined (for example, due to lack of connectivity) • "normal" - The resource has no known issues and can provide services in accordance with the interface specification • "warning" - The resource has known issues that do not prevent it providing services in accordance with the interface specification • "critical" - The resource has known issues that prevent it from providing services in accordance with the interface specification
healthStateDetails	array[type_astra_stateDetail]	True	JSON array of JSON objects that lists additional details about the current health. If no details are available, this array will be empty. • Minimum Items: 0

Name	Type	Required	Description
metadata	type_astra_metadata	True	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema

type_astra_metadata_list

Name	Type	Required	Description
continue	string	False	
count	number	False	
labels	array[type_astra_label]	False	
creationTimestamp	string	False	
modificationTimestamp	string	False	
createdBy	string	False	
modifiedBy	string	False	

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

Retrieve a volume for an app

GET `/accounts/{account_id}/k8s/v1/apps/{app_id}/volumes/{volume_id}`

Returns the stored Volume API resource in the JSON response body.

Parameters

Name	Type	In	Required	Description
account_id	string	path	True	ID of the containing account resource • example: {{.Account}}
app_id	string	path	True	ID of the containing app resource
volume_id	string	path	True	ID of the volume collection to list

Response

```
Status: 200, Returns the stored Volume API resource in the JSON response
body.
```

Name	Type	Required	Description
type	string	True	Media type of the resource. Defined values are: • "application/astra-volume"
version	string	True	Version of the resource. Defined values are: • "1.0" • "1.1" • "1.2"
id	string	True	Globally unique identifier of the resource. Defined values are: • Conforms to the UUIDv4 Schema

Name	Type	Required	Description
name	string	True	JSON string containing the name of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255
state	string	False	JSON string containing a value indicating the operational state of the volume.
size	string	False	JSON string containing the size of the volume.
used	number	False	JSON number containing the number of bytes of quota used by the provisioned volume.
usedPercentage	number	False	JSON number containing the percentage of volume used.
total	number	False	JSON number containing the total bytes of volumes provisioned.
creationToken	string	False	JSON string containing the creation token of the volume. Defined values are: • Minimum length: 1 • Maximum length: 255

Name	Type	Required	Description
snapshotPossible	string	False	JSON string indicating whether the driver can perform a snapshot of the volume. Defined values are: • "true" - Driver can take a snapshot of the volume • "false" - Driver cannot take a snapshot of the volume
storageClass	string	False	JSON string containing the storage class for the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255
pvcName	string	False	JSON string containing the PersistentVolumeClaim (PVC) name of the volume. • Defined values are: • Minimum length: 1 • Maximum length: 255
internalName	string	False	JSON string containing the internal name of the volume on the target storage cluster, if available. This name is obtained from the Kubernetes CSI driver. • Defined values are: • Minimum length: 1 • Maximum length: 255

Name	Type	Required	Description
appsUsing	array[string]	True	JSON array of JSON strings, each containing the ID of an application that is using the volume. Defined values are: • Conforms to the Astra Identifier Schema
storageBackendID	string	False	JSON string containing the ID of the storage backend that provides the volume. Defined values are: • Conforms to the Astra Identifier Schema
orchestrator	string	False	JSON string containing the orchestrator for this volume. This will be defined for Astra Data Store volumes. Defined values are: • "trident" - Trident has provisioned this volume • "vasa" - NetApp VASA Provider has provisioned this volume • "other" - The volume has been provisioned manually or the orchestrator is not known
serviceLevel	string	False	JSON string containing the quality of service (QoS) policy defined for this volume at the storage backend level. This will be defined for Astra Data Store volumes.

Name	Type	Required	Description
exportPolicy	string	False	JSON string containing the export policy associated with this volume. This will be defined for Astra Data Store volumes.
nodeId	string	False	JSON string containing the node ID serving the volume. This will be defined for Astra Data Store volumes. Defined values are: • Conforms to the Astra Identifier Schema
healthState	string	False	JSON string containing a value indicating the health state of the volume. This will be defined for Astra Data Store volumes. Defined values are: • "indeterminate" - The health of the resource cannot be determined (for example, due to lack of connectivity) • "normal" - The resource has no known issues and can provide services in accordance with the interface specification • "warning" - The resource has known issues that do not prevent it providing services in accordance with the interface specification • "critical" - The resource has known issues that prevent it from providing services in accordance with the interface specification

Name	Type	Required	Description
healthStateDetails	array[type_astra_stateDetail]	True	JSON array of JSON objects that lists additional details about the current health. If no details are available, this array will be empty. • Minimum Items: 0
metadata	type_astra_metadata	True	Client and service-specified metadata associated with the resource. Defined values are: • Conforms to the Astra Metadata Schema

Example response


```
{
  "type": "application/astra-volume",
  "version": "1.2",
  "id": "ab5c9fb8-ddba-526f-8bd3-8d5b72cf4f7f",
  "name": "myVolume",
  "state": "ready",
  "size": "512 GiB",
  "used": 10000000000,
  "usedPercentage": 10,
  "total": 10000000000,
  "creationToken": "ba7d4e0c-6ae3-4b6e-859d-9463e7ca4b4e",
  "snapshotPossible": "true",
  "storageClass": "gold",
  "pvcName": "myVolume",
  "internalName": "trident_sd234234_23424_897d_876sd65w",
  "appsUsing": [],
  "orchestrator": "trident",
  "serviceLevel": "gold",
  "exportPolicy": "astra-vp-default",
  "nodeId": "94d16dd2-9beb-4edb-ab58-1a00e9a90052",
  "healthState": "normal",
  "healthStateDetails": [],
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```


Response

```
Status: 401, Unauthorized
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	

Name	Type	Required	Description
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

Response

```
Status: 400, Bad request
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	
invalidParams	array[invalidParams]	False	List of invalid query parameters

Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

Response

```
Status: 403, Forbidden
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

Error

```
Status: 404, Not found
```

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
status	string	True	
correlationID	string	False	

Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't
found.",
  "status": "404"
}
```

Definitions

See Definitions

additionalDetails

type_astra_stateDetail

Name	Type	Required	Description
type	string	True	
title	string	True	
detail	string	True	
additionalDetails	additionalDetails	False	

type_astra_label

Name	Type	Required	Description
name	string	True	
value	string	True	

type_astra_metadata

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema

Name	Type	Required	Description
labels	array[type_astra_label]	True	
creationTimestamp	string	True	
modificationTimestamp	string	True	
createdBy	string	True	
modifiedBy	string	False	

invalidParams

Name	Type	Required	Description
name	string	True	Name of the invalid query parameter
reason	string	True	Reason why the query parameter is invalid

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